

JONATHAN ONG CHONG AN

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Availability: Jan 2027 – Jul 2027

Singapore, SG

EDUCATION

Bachelor of Science, Information Systems (Product Development)
Singapore Management University | Second Major: Computer Science (AI)

Jan 2023 – Dec 2027

SUMMARY

Production-experienced undergraduate skilled in PySpark, Hadoop ecosystem, MLOps pipeline delivery and DevOps Principles across cloud platforms. Passionate about distributed systems and building data infrastructure that scales.

EXPERIENCE

AI & Data Engineering Intern, Inland Revenue Authority of Singapore (IRAS)

May 2026 – Nov 2026

- Migrating legacy SAS data lake queries to PySpark across millions of production records, ensuring functional parity through rigorous DataFrame validation against SAS outputs.
- Optimising PySpark query performance and code structure for integration into a modernised data lake querying system.
- Enhancing CI/CD pipelines, Dockerisation, and workflow releases for an in-development MLOps codebase across UAT and production environments.
- Conducted adversarial testing and UAT evaluation for an internal GenAI LLM tool serving organisation-wide users.

DevOps Engineer Intern, YTL PowerSeraya

Jan 2026 – Apr 2026

- Built and rolled out Azure DevOps multi-stage YAML pipelines for 4 Azure applications (App Service & VM hosted), cutting deployment time from 60 to 20 minutes.
- Architected and implemented a self-hosted agent methodology to run IaaS CI/CD pipeline deployments within an internally isolated VNet on application VMs; documented the workflow for developer reference.
- Designed and built 2 Power BI dashboards for management weekly reporting, tracking KPIs across incident tickets, user stories, and project progress; featuring drill-down capabilities and real-time data connections.
- Facilitated internal Power BI learning sessions to drive adoption and awareness of self-service reporting capabilities.

Cloud Engineer Intern, Global Enterprise Mobility (GEM)

May 2025 – Aug 2025

- Provisioned AWS infrastructure using Terraform to improve environment parity and repeatability.
- Built 2 CI/CD pipelines with Jenkins (Groovy) integrating GCP Secret Manager for secure credential handling, reducing release time by 50% (40 to 20 minutes).
- Eliminated 100% of webhook duplicates by adding a DynamoDB idempotency layer, resolving a bug affecting 12–15% of transactions.

PROJECTS

MensWear (Microservices Web Application) | Team Lead

Aug 2025 – Nov 2025

- Led end-to-end deployment of a microservices e-commerce platform on AWS (ECS/Fargate, ALB) using Well-Architected principles; team awarded A grade.
- Designed a multi-database backend: DynamoDB (sessions), DocumentDB (catalog), and RDS (transactions) to align data stores with workload requirements.
- Deployed and operated 8+ microservices; maintained 99.9% uptime through robust service configuration.

YilongMa — RAG Data Pipeline

Aug 2025 – Dec 2025

- Built end-to-end RAG pipeline using ChromaDB, Llama 3.2 1B, and hybrid retrieval over Elon Musk transcript datasets.
- Transcribed and processed 50+ hours of audio via Whisper AI; built evaluation notebooks for retrieval quality and grounded response accuracy.

CERTIFICATIONS

Databricks Certified Associate Developer for Apache Spark | AWS Certified Cloud Practitioner

SKILLS

Languages: Python, SQL, Bash, JavaScript, Java

Big Data Tools/Frameworks: PySpark, Hadoop, Pandas, SAS (migration), Azure Synapse, PowerBI

Cloud: AWS, Azure (DevOps, Portal, Azure ML, Azure Synapse), Databricks

DevOps/IaC: Terraform, Docker, Jenkins, GitHub Actions, Azure Pipelines

Databases: Azure Data Lake, DynamoDB, PostgreSQL, MongoDB, RDS, Google Firestore